

Group & Ring Theory MATP33: Course Summary of Important Results

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Lecture 1

Theorem 1 (1.10). *Let G be a group and X a set. Then:*

- (i) *If X is a G -set, then the action of G on X induces a homomorphism $\varphi : G \rightarrow S_X$;*
- (ii) *Any homomorphism $\varphi : G \rightarrow S_X$ induces an action of G on X .*

Proof. (i) Define $\varphi : G \rightarrow S_X$ by $(\varphi(a))(x) = a \cdot x$ for $a \in G$ and $x \in X$. We must first check that $\varphi(a) \in S_X$ for each $a \in G$, i.e. that the function $x \mapsto a \cdot x$ is a bijection $X \rightarrow X$.

- *Injective:* Suppose $a \cdot x = a \cdot y$. Then $x = a^{-1} \cdot (a \cdot x) = a^{-1} \cdot (a \cdot y) = y$.
- *Surjective:* For any $x \in X$, we have $a \cdot (a^{-1} \cdot x) = x$, so $x \in \text{im}(\varphi(a))$.

Hence $\varphi(a) \in S_X$. To see φ is a homomorphism, let $a, b \in G$. Then for all $x \in X$,

$$(\varphi(ab))(x) = (ab) \cdot x = a \cdot (b \cdot x) = (\varphi(a))((\varphi(b))(x)) = (\varphi(a)\varphi(b))(x).$$

(ii) Define $a \cdot x := (\varphi(a))(x)$. Then $(ab) \cdot x = (\varphi(ab))(x) = (\varphi(a)\varphi(b))(x) = a \cdot (b \cdot x)$, and $e \cdot x = (\varphi(e))(x) = x$. Hence X is a G -set. $\square$

Theorem 2 (1.11 — Cayley's Theorem). *Every group G is isomorphic to a subgroup of the symmetric group S_G .*

Proof. By Theorem 1.10(i), left translation $a \cdot x := ax$ gives a homomorphism $\varphi : G \rightarrow S_G$. If $a \in \ker \varphi$, then $\varphi(a) = \text{id}_G$, so $ax = x$ for all $x \in G$, giving $a = e$. Hence $\ker \varphi = \{e\}$, and by the First Isomorphism Theorem $G \cong G / \ker \varphi \cong \text{im } \varphi \leq S_G$. $\square$

Definition 1 (Normal core). *Let G be a group and $H \leq G$. The normal core of H in G is*

$$\text{core}(H) := \bigcap_{x \in G} xHx^{-1}.$$

It is the largest normal subgroup of G contained in H .

Theorem 3 (1.12). *Let G be a group and $H \leq G$ a subgroup of index n . Then there is a homomorphism $\varphi : G \rightarrow S_n$ such that $\ker \varphi = \text{core}(H)$.*

Proof. Since $|G/H| = n$, we have $S_{G/H} \cong S_n$. By Theorem 1.10(i), the action of G on the coset space G/H by left multiplication gives a homomorphism $\varphi : G \rightarrow S_n$ with $\varphi(a)(xH) = axH$. Then

$$\begin{aligned} \ker \varphi &= \{a \in G \mid axH = xH \text{ for all } x \in G\} \\ &= \{a \in G \mid x^{-1}ax \in H \text{ for all } x \in G\} \\ &= \{a \in G \mid a \in xHx^{-1} \text{ for all } x \in G\} \\ &= \bigcap_{x \in G} xHx^{-1} = \text{core}(H). \end{aligned} \quad \square$$

Corollary 1 (1.13). *Let G be a group with a normal subgroup H of index n . Then G/H is isomorphic to a subgroup of S_n .*

Proof. Since $H \trianglelefteq G$, conjugation fixes H , so $\text{core}(H) = \bigcap_{x \in G} xHx^{-1} = H$. By Theorem 1.12, $\ker \varphi = H$, and by the First Isomorphism Theorem $G/H \cong \text{im } \varphi \leq S_n$. $\square$

Theorem 4 (1.24 — Orbit-Stabiliser Theorem). *Let G be a group acting on a set X . Then the set of all orbits in X under G is a partition of X . Furthermore, for each $x \in X$ there is a bijection $Gx \rightarrow G/G_x$, and hence*

$$|Gx| = |G : G_x|.$$

Therefore, if X is finite,

$$|X| = \sum_{x \in C} |G : G_x|,$$

where $C \subseteq X$ contains exactly one element from each orbit.

Proof. (Partition.) Define $x \sim y$ if $x = ay$ for some $a \in G$. This is an equivalence relation: reflexivity follows from $e \cdot x = x$, symmetry from $x = ay \Rightarrow y = a^{-1}x$, and transitivity from $x = ay, y = bz \Rightarrow x = (ab)z$. The equivalence class of x is exactly the orbit Gx , so the orbits partition X .

(Bijection $Gx \rightarrow G/G_x$.) Define

$$\varphi : Gx \rightarrow G/G_x, \quad ax \mapsto aG_x.$$

The key observation is the following chain of equivalences, for any $a, b \in G$:

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

- *Well-defined:* If $ax = bx$ (the same orbit element written using two different group elements), then $aG_x = bG_x$, so the coset φ assigns does not depend on the choice of representative.
- *Injective:* If $\varphi(ax) = \varphi(bx)$, i.e. $aG_x = bG_x$, then $ax = bx$ by the chain above.
- *Surjective:* Take any coset $aG_x \in G/G_x$. Since $a \in G$, we have $ax \in Gx$, and $\varphi(ax) = aG_x$.

Hence φ is a bijection, giving $|Gx| = |G/G_x| = |G : G_x|$. Summing over one representative per orbit gives $|X| = \sum_{x \in C} |G : G_x|$. $\square$

Theorem 5 (1.25 — Class Equation). *Let G be a finite group. Then the conjugacy classes partition G , and*

$$|G| = \sum_{a \in C} |G : C_G(a)|,$$

where $C \subseteq G$ contains exactly one element from each conjugacy class.

Proof. Let G act on itself by conjugation, i.e. $a \cdot x = axa^{-1}$. The orbit of x under this action is its conjugacy class $\text{ccl}(x) = \{axa^{-1} \mid a \in G\}$, and the stabiliser of x is $G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x)$. Applying Theorem 1.24 directly gives the partition into orbits and $|G| = \sum_{a \in C} |G : C_G(a)|$. $\square$